

COURSE CODE

SBT-DF202

COURSE TITLE

Basic Networking Skills for Digital Forensics

LAB TITLE

HTTP Analysis Using Wireshark - Text Traffic

NAME

Kumpon Emmanuel Nandom

STUDENT ID

2025/FWSD/11325

DATE

08/09/2026

PLATFORM USED

Kali Linux

INSTRUCTOR

Mr. Aminu Idris, AMCPN
Commandant

Lab Overview

You are a junior network forensic analyst asked to document a short plaintext HTTP session. You will create a local webpage containing your name and registration number, capture the traffic, reconstruct the TCP conversation, identify the HTTP request and response, and explain how each protocol layer contributes evidence.

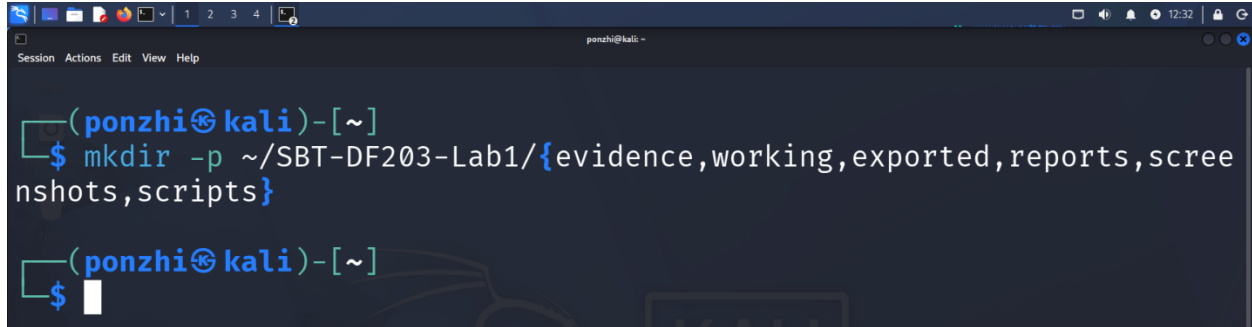
Create Your Evidence Folder Structure

```
mkdir -p ~/SBT-DF203-Lab1/{evidence,working,exported,reports,screenshots,scripts}
```

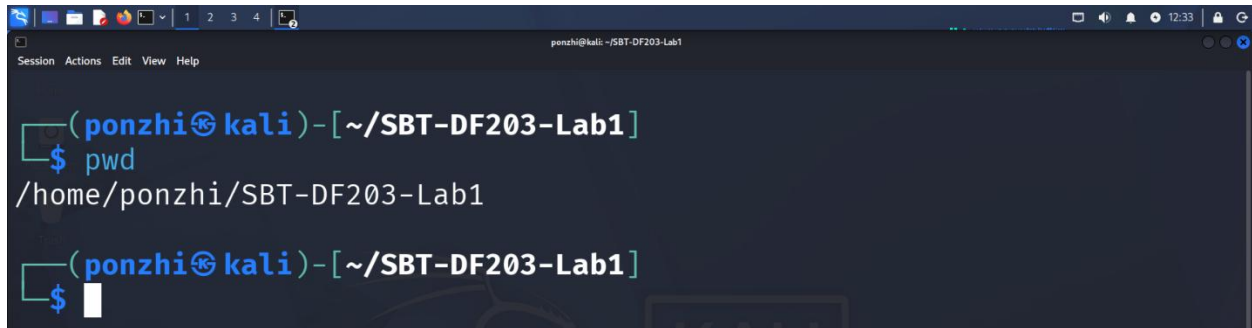
```
cd ~/SBT-DF203-Lab1
```

```
pwd
```

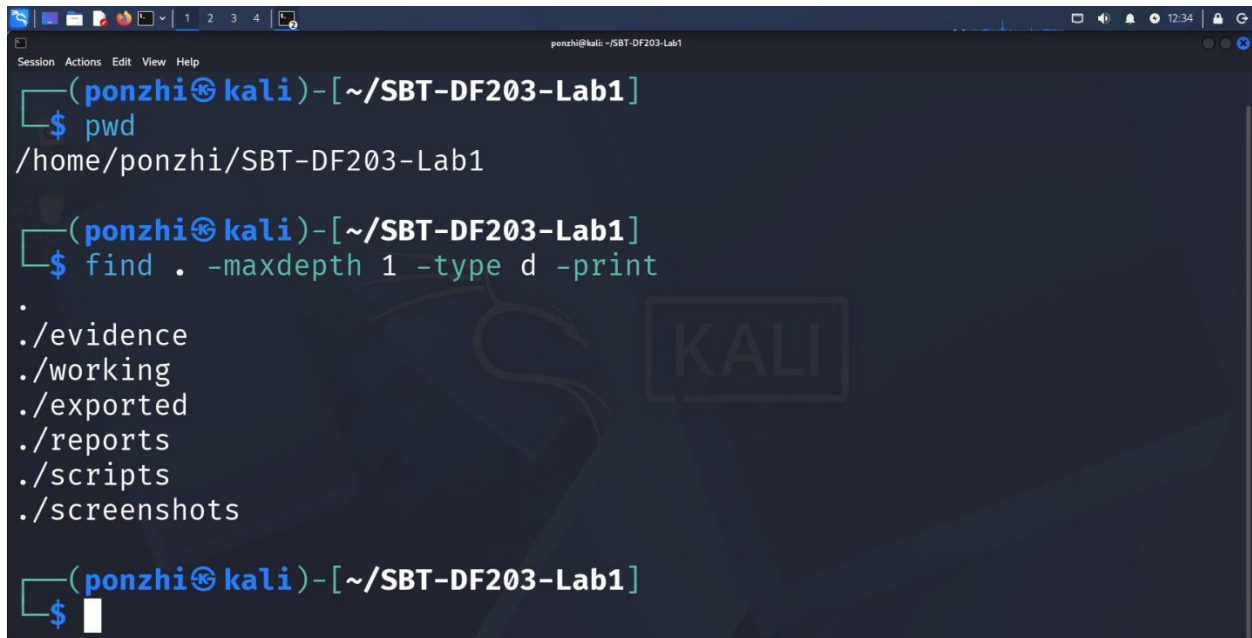
```
find . -maxdepth 1 -type d -print
```



```
ponzhi@kali: ~  
$ mkdir -p ~/SBT-DF203-Lab1/{evidence,working,exported,reports,screenshots,scripts}  
ponzhi@kali: ~  
$
```



```
ponzhi@kali: ~/SBT-DF203-Lab1  
$ pwd  
/home/ponzhi/SBT-DF203-Lab1  
ponzhi@kali: ~/SBT-DF203-Lab1  
$
```



```
ponzhi@kali: ~/SBT-DF203-Lab1  
$ pwd  
/home/ponzhi/SBT-DF203-Lab1  
ponzhi@kali: ~/SBT-DF203-Lab1  
$ find . -maxdepth 1 -type d -print  
.  
./evidence  
./working  
./exported  
./reports  
./scripts  
./screenshots  
ponzhi@kali: ~/SBT-DF203-Lab1  
$
```

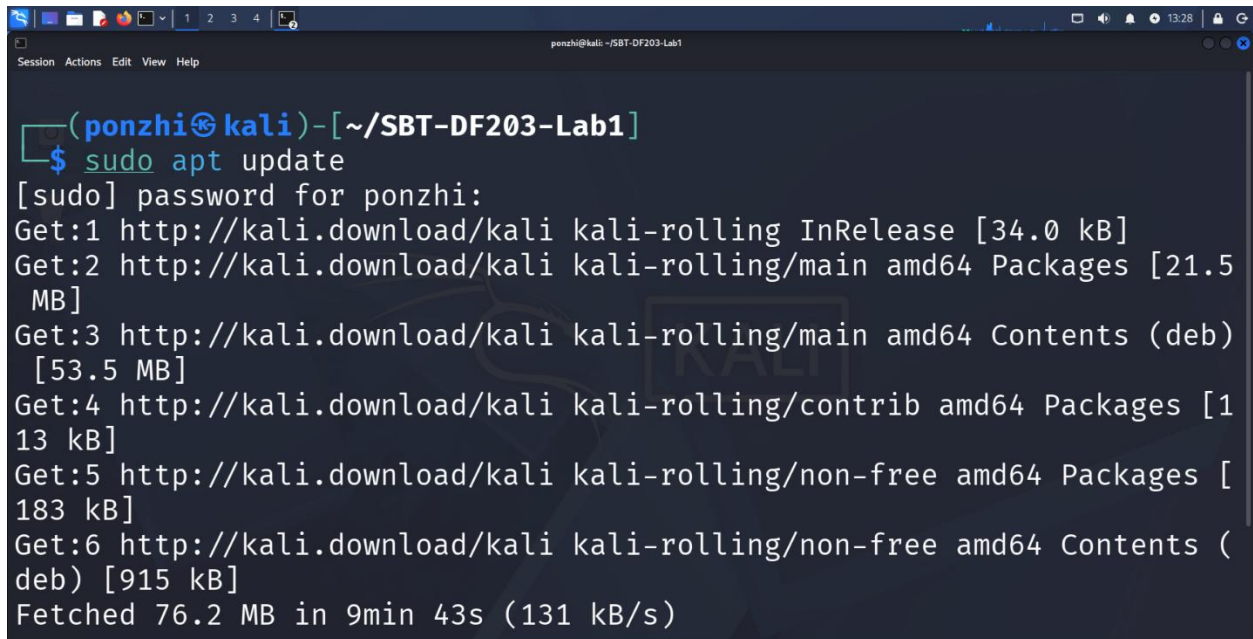
Install Tools and Create the Test Webpage

`sudo apt update`

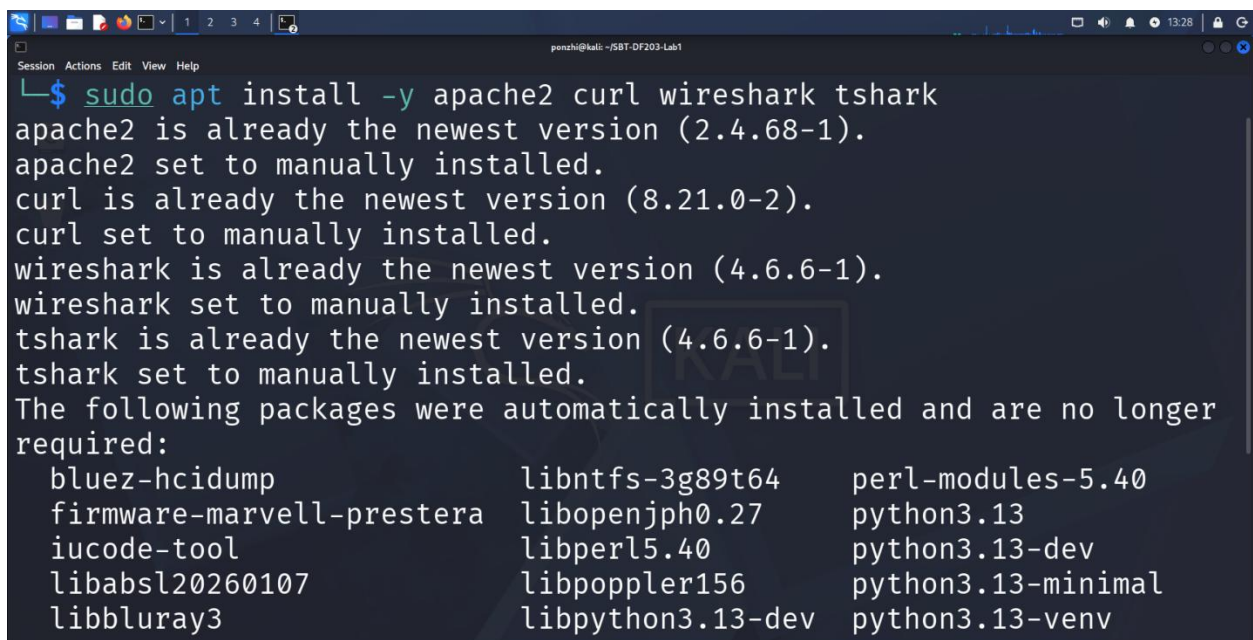
`sudo apt install -y apache2 curl wireshark tshark`

`sudo systemctl enable --now apache2`

```
printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name:
KUMPON EMMANUEL NANDOM</p><p>Reg No:
2025/FWSD/11325</p></body></html>\n' | sudo tee /var/www/html/basic.html
```

A terminal window titled 'ponzhi@kali: ~/SBT-DF203-Lab1' showing the output of the 'sudo apt update' command. The output lists updates for kali-rolling packages and their sizes. A large 'KALI' watermark is visible in the background.

```
(ponzhi@kali)-[~/SBT-DF203-Lab1]
$ sudo apt update
[sudo] password for ponzhi:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [13 kB]
Get:5 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 76.2 MB in 9min 43s (131 kB/s)
```

A terminal window titled 'ponzhi@kali: ~/SBT-DF203-Lab1' showing the output of the 'sudo apt install -y apache2 curl wireshark tshark' command. It reports that the specified packages are already the newest versions and lists other packages that were automatically installed. A large 'KALI' watermark is visible in the background.

```
$ sudo apt install -y apache2 curl wireshark tshark
apache2 is already the newest version (2.4.68-1).
apache2 set to manually installed.
curl is already the newest version (8.21.0-2).
curl set to manually installed.
wireshark is already the newest version (4.6.6-1).
wireshark set to manually installed.
tshark is already the newest version (4.6.6-1).
tshark set to manually installed.
The following packages were automatically installed and are no longer
required:
  bluez-hcidump      libntfs-3g89t64      perl-modules-5.40
  firmware-marvell-prestera libopenjph0.27      python3.13
  iucode-tool        libperl5.40         python3.13-dev
  libabsl20260107    libpoppler156       python3.13-minimal
  libbluray3         libpython3.13-dev   python3.13-venv
```

```

(ponzhi@kali)-[~/SBT-DF203-Lab1]
$ sudo systemctl enable --now apache2
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.service'.

(ponzhi@kali)-[~/SBT-DF203-Lab1]
$

(ponzhi@kali)-[~/SBT-DF203-Lab1]
$ sudo systemctl enable --now apache2
Created symlink '/etc/systemd/system/multi-user.target.wants/apache2.service' → '/usr/lib/systemd/system/apache2.service'.

(ponzhi@kali)-[~/SBT-DF203-Lab1]
$ printf '<!DOCTYPE html>\n<html><body><h1>SBT-DF203 HTTP Evidence</h1><p>Name: KUMPON EMMANUEL NANDOM</p><p>Reg No: 2025/FWSD/11325</p></body></html>\n' | sudo tee /var/www/html/basic.html

```

Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab1-KUMPON EMMANUEL NANDOM
Trainee name	KUMPON EMMANUEL NANDOM
Date and time started	08/09/2026 AND 6:20:56am
Evidence file name(s)	basic. pcapng
Source or generation method	Live capture using Wireshark on interface lo (filter: tcp port 80), while running curl against the locally hosted Apache page basic.html.
Original SHA-256	ee9c6d3c5d5be68939c52173834f2d9dfdbdd3ff2d7f3fd140a5b3ae9d98781c
Working-copy SHA-256	ee9c6d3c5d5be68939c52173834f2d9dfdbdd3ff2d7f3fd140a5b3ae9d98781c
Analysis workstation/VM	Kali/wireshark
Notes on any changes	

Analyze the TCP Three-Way Handshake

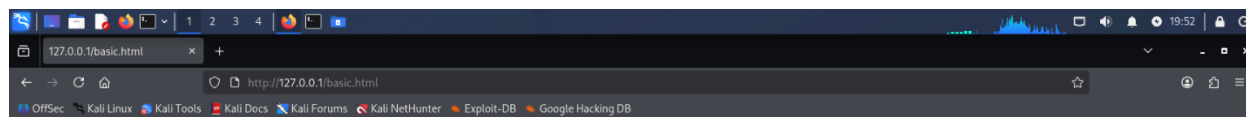
```
ponzhi@kali: ~/SBT-DF203-Lab1
Session Actions Edit View Help

(ponzhi@kali)-[~/SBT-DF203-Lab1]
$ sudo ss -lntp | grep ':80'
LISTEN 0      511          *:80          *:~ users:(("apache2",pid=67494,fd=4),("apache2",pid=67493,fd=4),("apache2",pid=67492,fd=4),("apache2",pid=67491,fd=4),("apache2",pid=67490,fd=4),("apache2",pid=67474,fd=4))

(ponzhi@kali)-[~/SBT-DF203-Lab1]
$
```

```
ponzhi@kali: ~/SBT-DF203-Lab1
Session Actions Edit View Help

(ponzhi@kali)-[~/SBT-DF203-Lab1]
$ curl -v http://127.0.0.1/basic.html 2>&1 | tee reports/curl_verbose.txt
* Trying 127.0.0.1:80 ...
* Established connection to 127.0.0.1 (127.0.0.1 port 80) from 127.0.0.1 port 60288
  % Total    % Received % Xferd  Average Speed   Time    Time     Time
  Current                                 Dload  Upload  Total   Spent    Left
  Speed
  0      0    0     0     0      0     0      0     0
    0* using HTTP/1.x
> GET /basic.html HTTP/1.1
> Host: 127.0.0.1
> User-Agent: curl/8.21.0
> Accept: */*
```



SBT-DF203 HTTP Evidence

Name: KUMPON EMMANUEL NANDOM

Reg No: 2025/PWSD/11325

basic.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 0

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	41642 → 80 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=1978654947 TSecr=0 WS=512
2	0.000048027	127.0.0.1	127.0.0.1	TCP	74	80 → 41642 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=1381968452 TSecr=1978654947 WS=512
3	0.000098229	127.0.0.1	127.0.0.1	TCP	66	41642 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1978654948 TSecr=1381968452
4	0.000921638	127.0.0.1	127.0.0.1	HTTP	149	GET /basic.html HTTP/1.1
5	0.001038543	127.0.0.1	127.0.0.1	TCP	66	80 → 41642 [ACK] Seq=1 Ack=84 Win=65536 Len=0 TSval=1381968452 TSecr=1978654948
6	0.001702171	127.0.0.1	127.0.0.1	HTTP	457	HTTP/1.1 200 OK (text/html)
7	0.001759273	127.0.0.1	127.0.0.1	TCP	66	41642 → 80 [ACK] Seq=84 Ack=392 Win=65536 Len=0 TSval=1978654949 TSecr=1381968453
8	0.002205691	127.0.0.1	127.0.0.1	TCP	66	41642 → 80 [FIN, ACK] Seq=84 Ack=392 Win=65536 Len=0 TSval=1978654949 TSecr=1381968453
9	0.002348897	127.0.0.1	127.0.0.1	TCP	66	80 → 41642 [FIN, ACK] Seq=392 Ack=85 Win=65536 Len=0 TSval=1381968454 TSecr=1978654949
10	0.002396699	127.0.0.1	127.0.0.1	TCP	66	41642 → 80 [ACK] Seq=85 Ack=393 Win=65536 Len=0 TSval=1978654950 TSecr=1381968454

Frame 1: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 41642, Dst Port: 80, Seq: 0, Len: 0

Source Port: 41642

Destination Port: 80

[Stream index: 0]

[Stream Packet Number: 1]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 4197533495

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 0

Acknowledgment number (raw): 0

1010 = Header Length: 40 bytes (10)

Flags: 0x02 (SYN)

Window: 65495

[Calculated window size: 65495]

Checksum: 0xfe30 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), [Timestamps]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

basic.pcapng

Packets: 14 · Displayed: 10 (71.4%) · Dropped: 0 (0.0%)

Profile: Default

basic.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.stream eq 0

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	127.0.0.1	127.0.0.1	TCP	74	41642 → 80 [SYN] Seq=0 Win=65495 Len=0 MSS=65495 SACK_PERM TSval=1978654947 TSecr=0 WS=512
2	0.000048027	127.0.0.1	127.0.0.1	TCP	74	80 → 41642 [SYN, ACK] Seq=0 Ack=1 Win=65483 Len=0 MSS=65495 SACK_PERM TSval=1381968452 TSecr=1978654947 WS=512
3	0.000098229	127.0.0.1	127.0.0.1	TCP	66	41642 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0 TSval=1978654948 TSecr=1381968452
4	0.000921638	127.0.0.1	127.0.0.1	HTTP	149	GET /basic.html HTTP/1.1
5	0.001038543	127.0.0.1	127.0.0.1	TCP	66	80 → 41642 [ACK] Seq=1 Ack=84 Win=65536 Len=0 TSval=1381968452 TSecr=1978654948
6	0.001702171	127.0.0.1	127.0.0.1	HTTP	457	HTTP/1.1 200 OK (text/html)
7	0.001759273	127.0.0.1	127.0.0.1	TCP	66	41642 → 80 [ACK] Seq=84 Ack=392 Win=65536 Len=0 TSval=1978654949 TSecr=1381968453
8	0.002205691	127.0.0.1	127.0.0.1	TCP	66	41642 → 80 [FIN, ACK] Seq=84 Ack=392 Win=65536 Len=0 TSval=1978654949 TSecr=1381968453
9	0.002348897	127.0.0.1	127.0.0.1	TCP	66	80 → 41642 [FIN, ACK] Seq=392 Ack=85 Win=65536 Len=0 TSval=1381968454 TSecr=1978654949
10	0.002396699	127.0.0.1	127.0.0.1	TCP	66	41642 → 80 [ACK] Seq=85 Ack=393 Win=65536 Len=0 TSval=1978654950 TSecr=1381968454

Frame 2: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)

Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1

Transmission Control Protocol, Src Port: 80, Dst Port: 41642, Seq: 0, Ack: 1, Len: 0

Source Port: 80

Destination Port: 41642

[Stream index: 0]

[Stream Packet Number: 2]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 3101523320

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 4197533496

1010 = Header Length: 40 bytes (10)

Flags: 0x012 (SYN, ACK)

Window: 65483

[Calculated window size: 65483]

Checksum: 0xfe30 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), [Timestamps]

[SEQ/ACK analysis]

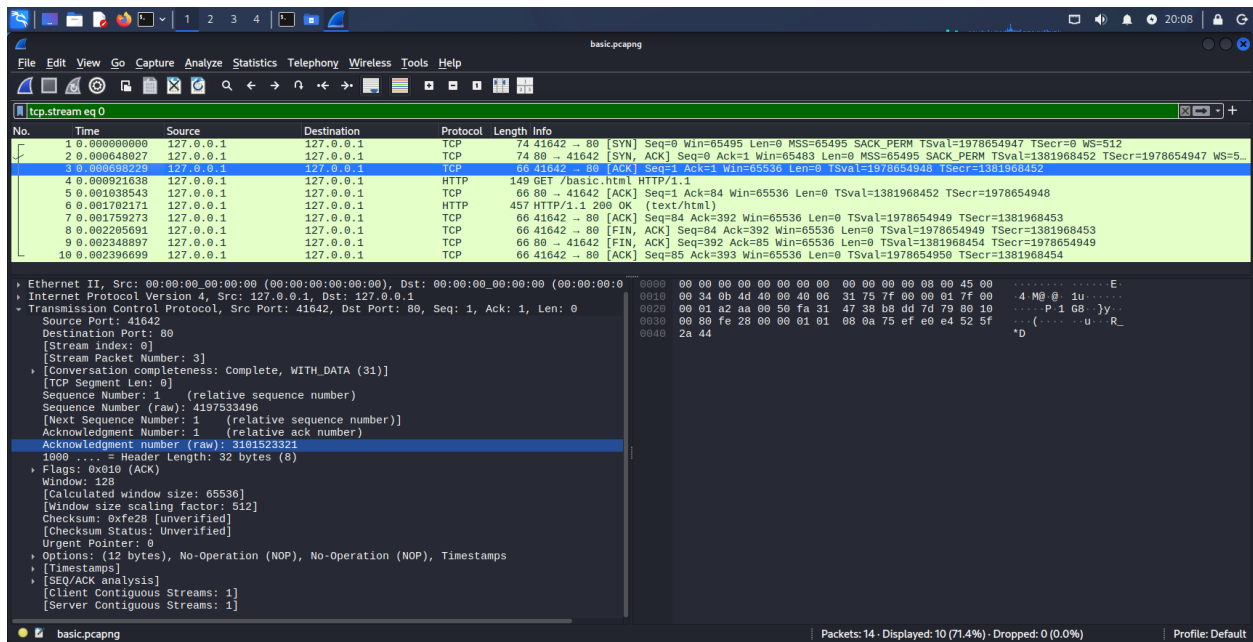
[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

basic.pcapng

Packets: 14 · Displayed: 10 (71.4%) · Dropped: 0 (0.0%)

Profile: Default



Open the working copy and apply the display filter tcp.stream eq 0. Identify the first three packets and complete the table.

Packet	Flags	Client Seq	Server Seq/ACK	Timestamp	Interpretation
1	SYN	Seq=0 (raw 4197533495)	-	0.000000000	Client requests a TCP session.
2	SYN, ACK	-	Seq=0 (raw 3101523320), Ack=1 (raw 4197533496)	0.000648027	Server acknowledges and provides its initial sequence number.
3	ACK	Seq=1 (raw 4197533496), Ack=1 (raw 3101523321)	-	0.000698229	Client acknowledges; the connection becomes established.

Examine the HTTP Request and Response

Filter: http.request → click the GET packet → expand Hypertext Transfer Protocol in the details pane to read the method, URI, Host, and User-Agent.

The screenshot shows the Wireshark interface with a packet capture named 'basic.pcapng'. The packet list pane shows a single packet (No. 4) at time 0.000921638, source 127.0.0.1, destination 127.0.0.1, protocol HTTP, length 149, and info GET /basic.html HTTP/1.1. The packet details pane shows the following structure:

- Frame 4: Packet, 149 bytes on wire (1192 bits), 149 bytes captured (1192 bits) on interface
- Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Transmission Control Protocol, Src Port: 41642, Dst Port: 80, Seq: 1, Ack: 1, Len: 83
- Hypertext Transfer Protocol

The packet bytes pane shows the raw data in hexadecimal and ASCII. The ASCII column shows the following text:

```
.....E
..N0.0.11.....
....P1G8..}y..
...{.....u..R..
*GET /b asic.htm
l HTTP/1.1. Host
: 127.0.0.1 Use
r-Agent: curl/8.
21.0. Ac cept: */
*..
```

The screenshot shows the Wireshark interface with the same packet capture. The packet list pane shows the same packet (No. 4). The packet details pane shows the following structure:

- Frame 4: Packet, 149 bytes on wire (1192 bits), 149 bytes captured (1192 bits) on interface
- Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Transmission Control Protocol, Src Port: 41642, Dst Port: 80, Seq: 1, Ack: 1, Len: 83
- Hypertext Transfer Protocol
 - GET /basic.html HTTP/1.1\r\n
 - Host: 127.0.0.1\r\n
 - User-Agent: curl/8.21.0\r\n
 - Accept: */*\r\n
 - \r\n
 - [Response in frame: 6]
 - [Full request URI: http://127.0.0.1/basic.html]

The packet bytes pane shows the raw data in hexadecimal and ASCII. The ASCII column shows the following text:

```
.....E
..N0.0.11.....
....P1G8..}y..
...{.....u..R..
*GET /b asic.htm
l HTTP/1.1. Host
: 127.0.0.1 Use
r-Agent: curl/8.
21.0. Ac cept: */
*..
```

basic.pcapng

2026
🔒

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

🔍
🔍
🔍
🔍
🔍
🔍
🔍
🔍

http.response
🔍

No.	Time	Source	Destination	Protocol	Length	Info
6	0.001702171	127.0.0.1	127.0.0.1	HTTP	457	HTTP/1.1 200 OK (text/html)

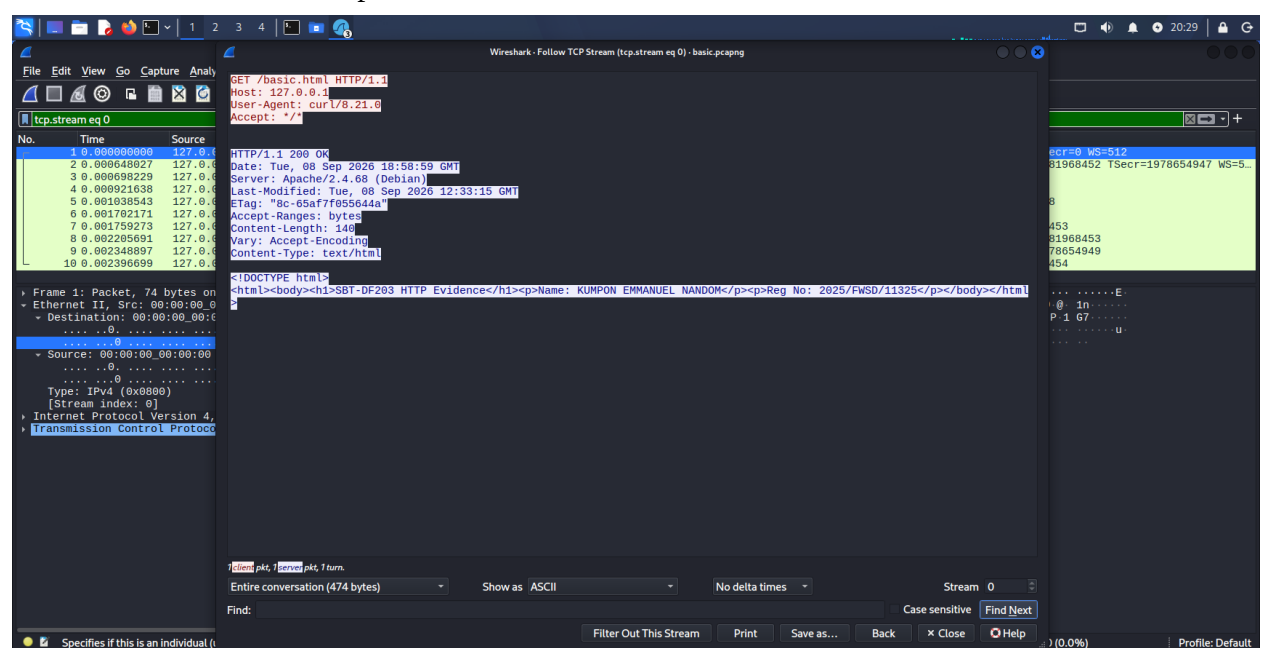
```

> Frame 6: Packet, 457 bytes on wire (3656 bits), 457 bytes captured (3656 bits) on interface
> Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 80, Dst Port: 41642, Seq: 1, Ack: 84, Len: 391
> Hypertext Transfer Protocol
  > HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Date: Tue, 08 Sep 2026 18:58:59 GMT\r\n
    Server: Apache/2.4.68 (Debian)\r\n
    Last-Modified: Tue, 08 Sep 2026 12:33:15 GMT\r\n
    ETag: "8c-65af7f055644a"\r\n
    Accept-Ranges: bytes\r\n
    Content-Length: 140\r\n
    Vary: Accept-Encoding\r\n
    Content-Type: text/html\r\n
    \r\n
    [request in frame 4]
    [Time since request: 780.533 microseconds]
    [request URI: /basic.html]
    [full request URI: http://127.0.0.1/basic.html]
    File Data: 140 bytes
  > Line-based text data: text/html (2 lines)

```

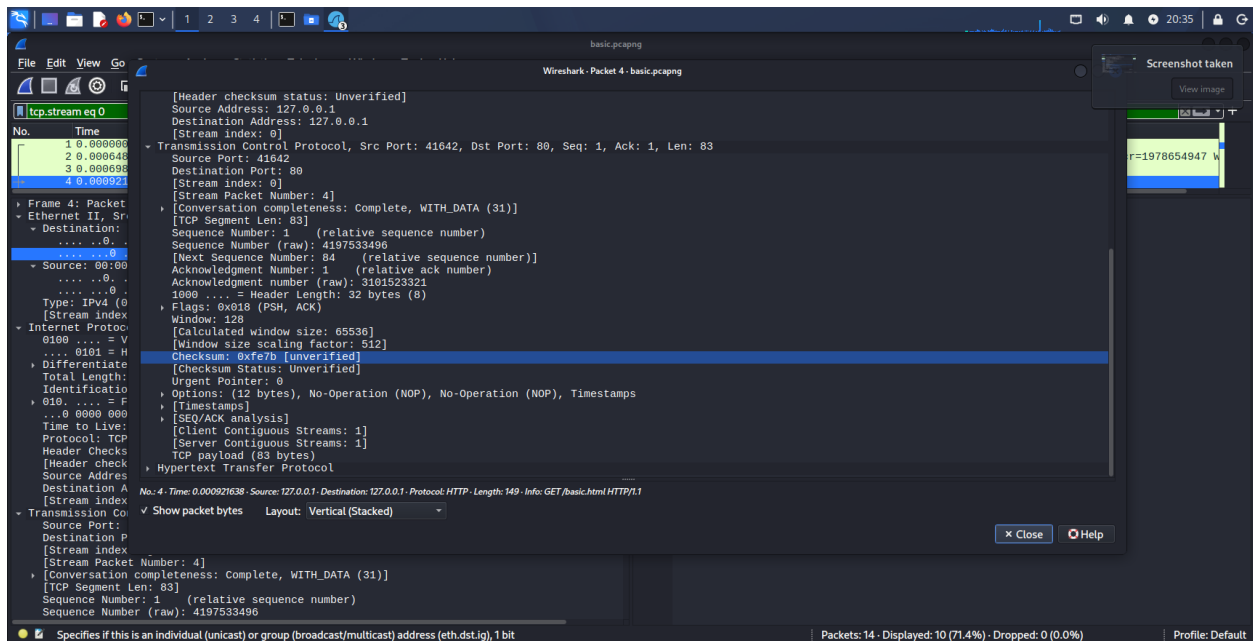
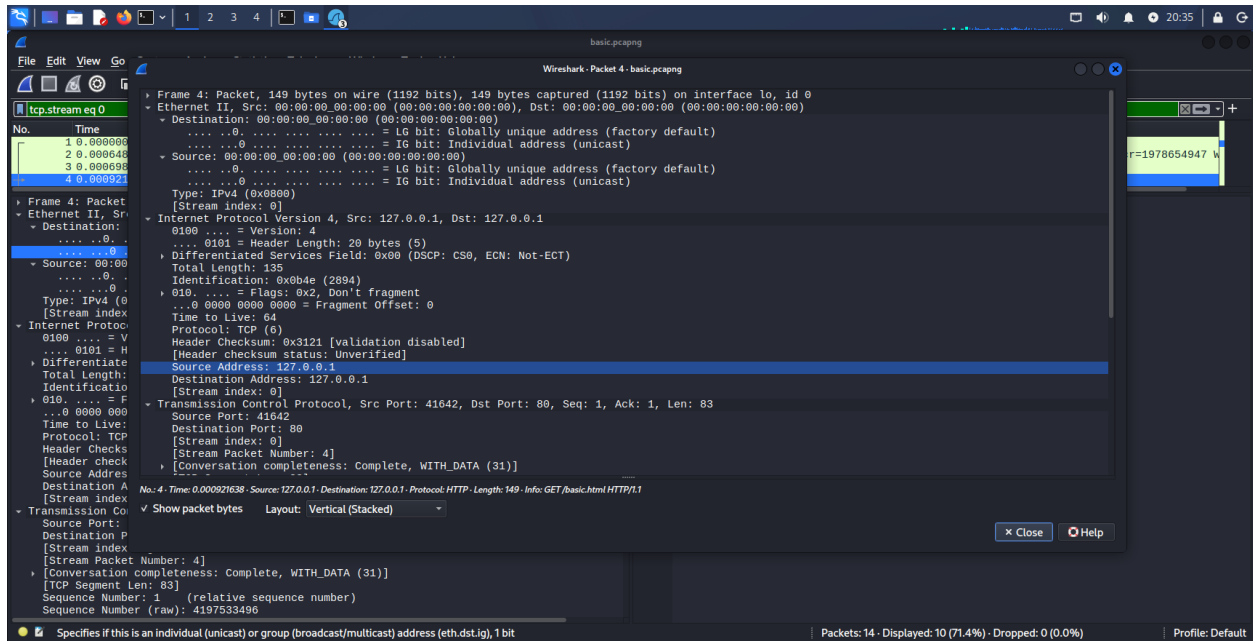
basic.pcapng
Packets: 14 - Displayed: 1 (7.1%) - Dropped: 0 (0.0%)
Profile: Default

Compare the TCP payload length (shown in the TCP layer) against how the sequence/ack numbers increase between packets.



Encapsulation and Connection Closure

Click the HTTP GET packet and expand, top to bottom, in the packet details pane: Frame, Ethernet II (if present), Internet Protocol Version 4, Transmission Control Protocol, Hypertext Transfer Protocol.



Filter: tcp.flags.fin==1 || tcp.flags.reset==1 to find the closing packets, and note the FIN/ACK or RST sequence.

The screenshot shows the Wireshark interface with a packet capture filter of `tcp.flags.fin==1 || tcp.flags.reset==1`. The packet list shows a TCP packet (No. 9) from 127.0.0.1 to 127.0.0.1, Seq=392, Ack=85, Win=65536, Len=0, TSval=1381968454, TSecr=1978654949. The packet details pane shows the following information:

- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Version: 4
- Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 52
- Identification: 0xb50 (2896)
- Flags: 0x2, Don't fragment
- Fragment Offset: 0
- Time to Live: 64
- Protocol: TCP (6)
- Header Checksum: 0x3172 [validation disabled]
- Source Address: 127.0.0.1
- Destination Address: 127.0.0.1
- Transmission Control Protocol, Src Port: 41642, Dst Port: 80, Seq: 84, Ack: 392, Len: 0
- Source Port: 41642
- Destination Port: 80
- Stream index: 0
- Stream Packet Number: 8
- Conversation completeness: Complete, WITH_DATA (31)
- TCP Segment Len: 0
- Sequence Number: 84 (relative sequence number)
- Sequence Number (raw): 4197533579
- Next Sequence Number: 85 (relative sequence number)
- Acknowledgment Number: 392 (relative ack number)
- Acknowledgment number (raw): 3101523712
- Header Length: 32 bytes (8)
- Flags: 0x011 (FIN, ACK)

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol header, and TCP header.

The screenshot shows the Wireshark interface with the same packet capture filter. The packet list shows a TCP packet (No. 9) from 127.0.0.1 to 127.0.0.1, Seq=392, Ack=85, Win=65536, Len=0, TSval=1381968454, TSecr=1978654949. The packet details pane shows the following information:

- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Version: 4
- Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 52
- Identification: 0xb50 (2896)
- Flags: 0x2, Don't fragment
- Fragment Offset: 0
- Time to Live: 64
- Protocol: TCP (6)
- Header Checksum: 0x3172 [validation disabled]
- Source Address: 127.0.0.1
- Destination Address: 127.0.0.1
- Transmission Control Protocol, Src Port: 41642, Dst Port: 80, Seq: 84, Ack: 392, Len: 0
- Source Port: 41642
- Destination Port: 80
- Stream index: 0
- Stream Packet Number: 8
- Conversation completeness: Complete, WITH_DATA (31)
- TCP Segment Len: 0
- Sequence Number: 84 (relative sequence number)
- Sequence Number (raw): 4197533579
- Next Sequence Number: 85 (relative sequence number)
- Acknowledgment Number: 392 (relative ack number)
- Acknowledgment number (raw): 3101523712
- Header Length: 32 bytes (8)
- Flags: 0x011 (FIN, ACK)
- Window: 128
- Calculated window size: 65536
- Window size scaling factor: 512
- Checksum: 0xfe20 [unverified]
- Checksum Status: Unverified
- Urgent Pointer: 0
- Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
- Timestamps
- Client Contiguous Streams: 1
- Server Contiguous Streams: 1

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol header, and TCP header.

Required Forensic Findings

Finding	Value/Observation
Capture start and end time	Relative: 0.000000000 s to 0.002396699 s. Date Tue, 08 Sep 2026 18:58:59 GMT.
Client IP and port	127.0.0.1 : 41642
Server IP and port	127.0.0.1 : 80
Client initial sequence number	Relative Seq = 0 (raw 4197533495) - SYN
Server initial sequence number	Relative Seq = 0 (raw 3101523320) - SYN-ACK
HTTP request timestamp	0.000921638 s
Requested URI	http://127.0.0.1/basic.html
HTTP response code	200 OK
Content type and length	text/html, Content-Length: 140 bytes
Connection termination method	4-way FIN close, client [FIN, ACK] Seq=84 Ack=392 - server [FIN, ACK] Seq=392 Ack=85 - client final [ACK].
MAC-address observation	Both source and destination Ethernet addresses are 00:00:00_00:00:00 as expected on the lo loopback interface, which has no real NIC/MAC.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.